实验 1.1

下载 hello-world 镜像

```
root@cg:~# docker pull hello-world:latest
latest: Pulling from library/hello-world
719385e32844: Pull complete
Digest: sha256:4f53e2564790c8e7856ec08e384732aa38dc43c52f02952483e3f003afbf23db
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
root@cg:~#
```

使用 docker run 在容器内运行应用程序，输出 Hello world

```
root@cg:~# docker run hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

root@cg:~#
```

使用 Ubuntu:16.04 镜像生成容器，并输出 hello world

```
root@cg:~# docker pull ubuntu:16.04
16.04: Pulling from library/ubuntu
58690f9b18fc: Pull complete
b51569e7c507: Pull complete
da8ef40b9eca: Pull complete
fb15d46c38dc: Pull complete
Digest: sha256:1f1a2d56de1d604801a9671f301190704c25d604a416f59e03c04f5c6ffee0d6
Status: Downloaded newer image for ubuntu:16.04
docker.io/library/ubuntu:16.04
root@cg:~#
root@cg:~# docker run ubuntu:16.04 /bin/echo "hello world"
hello world
root@cg:~#
```

运行交互式容器

```
root@cg:~# docker run -i -t ubuntu:16.04 /bin/bash
root@ca333e8b5076:/# cat /proc/version
Linux version 5.4.0-150-generic (buildd@bos03-amd64-012) (gcc version 7.5.0 (Ubu
ntu 7.5.0-3ubuntu1~18.04)) #167~18.04.1-Ubuntu SMP Wed May 24 00:51:42 UTC 2023
root@ca333e8b5076:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
root@ca333e8b5076:/# exit
exit
root@cg:~#
```

后台方式启动容器

启动容器

```
root@cg:~# docker run -d ubuntu:16.04 /bin/bash -c "while true;do echo hello wor
ld; sleep 1; done"
968e86373512edcdf188d63fe55e7a759a25ffb8af54fee0706cf98da0824565
root@cg:~#
```

查看容器

```
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
       NAMES
968e86373512        ubuntu:16.04        "/bin/bash -c 'while…"   About a minute ago   Up About a minute
       confident_dewdney
root@cg:~#
```

查看输出

```
root@cg:~# docker ps
CONTAINER ID        IMAGE
       NAMES
968e86373512        ubuntu:16.04
       confident_dewdney
root@cg:~# docker logs 968e86373512
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
```

```
root@cg:~# docker logs confident_dewdney
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
```

停止容器

```
root@cg:~# docker stop 968e

968e
root@cg:~#
root@cg:~# docker 啥
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
 NAMES
root@cg:~#
```

```
root@cg:~# docker stop confident_dewdney
confident_dewdney
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
 NAMES
root@cg:~#
```

实验 1.2

查看所有命令

```
root@cg:~# docker

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Options:
      --config string      Location of client config files (default
                           "/root/.docker")
  -c, --context string     Name of the context to use to connect to the
                           daemon (overrides DOCKER_HOST env var and
                           default context set with "docker context use")
  -D, --debug              Enable debug mode
  -H, --host list          Daemon socket(s) to connect to
  -l, --log-level string   Set the logging level
                           ("debug"|"info"|"warn"|"error"|"fatal")
                           (default "info")
      --tls                Use TLS; implied by --tlsverify
      --tlscacert string   Trust certs signed only by this CA (default
                           "/root/.docker/ca.pem")
      --tlscert string     Path to TLS certificate file (default
                           "/root/.docker/cert.pem")
      --tlskey string      Path to TLS key file (default
                           "/root/.docker/key.pem")
```

```
root@cg:~# docker run --help

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Run a command in a new container

Options:
      --add-host list                  Add a custom host-to-IP mapping (host:ip)
  -a, --attach list                    Attach to STDIN, STDOUT or STDERR
      --blkio-weight uint16            Block IO (relative weight), between 10 and 1000, or 0 to disable (default 0)
      --blkio-weight-device list       Block IO weight (relative device weight) (default [])
      --cap-add list                   Add Linux capabilities
      --cap-drop list                  Drop Linux capabilities
      --cgroup-parent string           Optional parent cgroup for the container
      --cidfile string                 Write the container ID to the file
      --cpu-period int                 Limit CPU CFS (Completely Fair Scheduler) period
      --cpu-quota int                  Limit CPU CFS (Completely Fair Scheduler) quota
      --cpu-rt-period int              Limit CPU real-time period in microseconds
      --cpu-rt-runtime int             Limit CPU real-time runtime in microseconds
  -c, --cpu-shares int                 CPU shares (relative weight)
      --cpus decimal                   Number of CPUs
      --cpuset-cpus string             CPUs in which to allow execution (0-3, 0,1)
      --cpuset-mems string             MEMs in which to allow execution (0-3, 0,1)
  -d, --detach                         Run container in background and print container ID
      --detach-keys string             Override the key sequence for detaching a container
      --device list                    Add a host device to the container
      --device-cgroup-rule list        Add a rule to the cgroup allowed devices list
      --device-read-bps list           Limit read rate (bytes per second) from a device (default [])
      --device-read-iops list          Limit read rate (IO per second) from a device (default [])
      --device-write-bps list          Limit write rate (bytes per second) to a device (default [])
      --device-write-iops list         Limit write rate (IO per second) to a device (default [])
      --disable-content-trust          Skip image verification (default true)
      --dns list                       Set custom DNS servers
      --dns-option list                Set DNS options
      --dns-search list                Set custom DNS search domains
      --domainname string              Container NIS domain name
      --entrypoint string              Overwrite the default ENTRYPOINT of the image
  -e, --env list                       Set environment variables
      --env-file list                  Read in a file of environment variables
      --expose list                    Expose a port or a range of ports
      --gpus gpu-request               GPU devices to add to the container ('all' to pass all GPUs)
      --group-add list                 Add additional groups to join
```

获取镜像

```
root@cg:~# docker load < /headless/Desktop/images/ubuntu_latest.tar.gz
ccdbb30308cc: Loading layer [==================================================>]  75.07MB/75.07MB
63c99163f472: Loading layer [==================================================>]  15.36kB/15.36kB
2f140462f3bc: Loading layer [==================================================>]  3.584kB/3.584kB
Loaded image: ubuntu:latest
root@cg:~# docker run -it ubuntu /bin/bash
root@0974dcc21d76:/#
```

启动、停止容器

```
root@cg:~# docker run -it ubuntu /bin/bash
root@0974dcc21d76:/# exit
exit
root@cg:~# docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED              STATUS                      PORTS
         NAMES
0974dcc21d76        ubuntu              "/bin/bash"         About a minute ago   Exited (0) 6 seconds ago
         hardcore_shannon
root@cg:~#
root@cg:~# docker start 0974
0974
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED              STATUS              PORTS
  NAMES
0974dcc21d76        ubuntu              "/bin/bash"         About a minute ago   Up 5 seconds
  hardcore_shannon
root@cg:~#
```

后台运行，停止，重启

```
root@cg:~# docker run -itd --name ubuntu_test ubuntu /bin/bash
d65655b44862eb4bb01bdce7a1939e47967824ba47334eaabc86ebc94f07b925
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
 NAMES
d65655b44862        ubuntu              "/bin/bash"         15 seconds ago      Up 13 seconds
 ubuntu_test
root@cg:~# docker stop d656
d656
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
 NAMES
root@cg:~# docker restart d656
d656
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
 NAMES
d65655b44862        ubuntu              "/bin/bash"         55 seconds ago      Up 3 seconds
 ubuntu_test
root@cg:~#
```

进入容器

```
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS              PORTS               NAMES
d65655b44862        ubuntu              "/bin/bash"         6 minutes ago
Up 22 seconds                           ubuntu_test
root@cg:~# docker exec d656
"docker exec" requires at least 2 arguments.
See 'docker exec --help'.

Usage:  docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

Run a command in a running container
root@cg:~# docker exec -it d656 /bin/bash
root@d65655b44862:/#
```

导出容器

导入容器

```
root@cg:~/Desktop# docker import ubuntu_test.tar cg/my_ubuntu:v1.0
sha256:583534f3240d7d7e12a1e01f216f53bba84bcfe492e3f8f39f5d14c265fc2736
root@cg:~/Desktop# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED
SIZE
cg/my_ubuntu        v1.0                583534f3240d        5 seconds ago
72.7MB
ubuntu              latest              7e0aa2d69a15        2 years ago
72.7MB
```

删除容器

```
root@cg:~/Desktop# docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS                      PORTS               NAMES
d65655b44862        ubuntu              "/bin/bash"         15 minutes ago
Up 8 minutes                                    ubuntu_test
0974dcc21d76        ubuntu              "/bin/bash"         18 minutes ago
Exited (0) 15 minutes ago                       hardcore_shannon
root@cg:~/Desktop# docker stop d656
d656
root@cg:~/Desktop# docker rm d656
d656
root@cg:~/Desktop# docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS                      PORTS               NAMES
0974dcc21d76        ubuntu              "/bin/bash"         18 minutes ago
Exited (0) 15 minutes ago                       hardcore_shannon
root@cg:~/Desktop# docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
0974dcc21d76e214f29c7ddb2f79fcf151d07671341d77ca3ffdd780da783b52

Total reclaimed space: 5B
root@cg:~/Desktop#
```

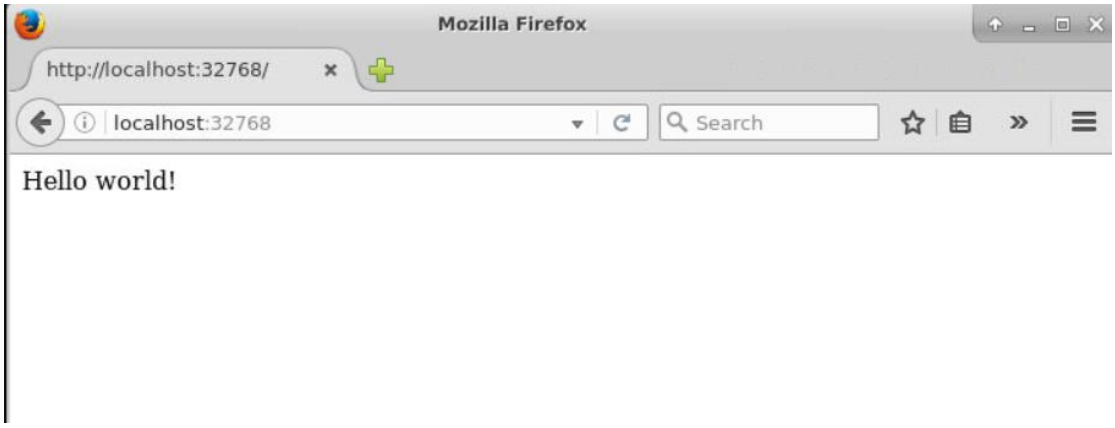
环境准备

```
USER_ID: 0, GROUP_ID: 0
root@cg:~# cd course/docker/webapp/webapp
root@cg:~/course/docker/webapp/webapp# pwd
/headless/course/docker/webapp/webapp
root@cg:~/course/docker/webapp/webapp# ls
0.0.0.0:5000  app.py  Procfile  requirements.txt  tests.py
root@cg:~/course/docker/webapp/webapp#
```

```
root@cg:~/course/docker/webapp/webapp# docker load < /headless/Desktop/images/tr
aining_webapp_latest.tar.gz
1154ba695078: Loading layer  197.2MB/197.2MB
528c8710fd95: Loading layer  208.9kB/208.9kB
37ee47034d9b: Loading layer  4.608kB/4.608kB
5f70bf18a086: Loading layer  1.024kB/1.024kB
b75c0703b86b: Loading layer  20.82MB/20.82MB
5c121779bb29: Loading layer  141.2MB/141.2MB
3ee91c5cb95b: Loading layer   2.56kB/2.56kB
6bbb1d0f8452: Loading layer  4.685MB/4.685MB
b44b0832d4c6: Loading layer  7.168kB/7.168kB
Loaded image: training/webapp:latest
```

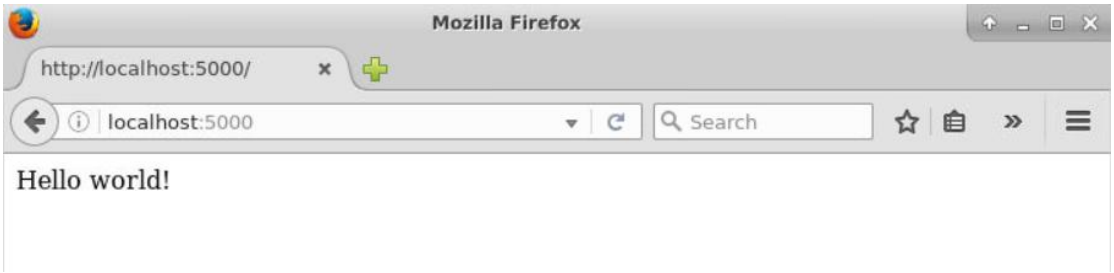
查看代码并使用 webapp 容器运行应用

```
root@cg:~/course/docker/webapp/webapp# cat app.py
import os

from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    provider = str(os.environ.get('PROVIDER', 'world'))
    return 'Hello '+provider+'!'

if __name__ == '__main__':
    # Bind to PORT if defined, otherwise default to 5000.
    port = int(os.environ.get('PORT', 5000))
    app.run(host='0.0.0.0', port=port)
root@cg:~/course/docker/webapp/webapp# docker run -d -P training/webapp python a
pp.py
3dd7c49b5f0861577b9abe3076d37476f9d34429582c8940bd65e93b467e4e29
root@cg:~/course/docker/webapp/webapp# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS              PORTS                    NAMES
3dd7c49b5f08        training/webapp     "python app.py"     9 seconds ago
Up 8 seconds        0.0.0.0:32768->5000/tcp   interesting_hawking
root@cg:~/course/docker/webapp/webapp#
```



指定端口运行

```
root@cg:~/course/docker/webapp/webapp# docker run -d -p 5000:5000 training/webapp python app.p
y
22cc30135359b35b818d5791d449f29118739b68ed2bd8332d1344b68b17dea3
root@cg:~/course/docker/webapp/webapp# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
      PORTS                    NAMES
22cc30135359        training/webapp     "python app.py"     5 seconds ago       Up 4 seconds
      0.0.0.0:5000->5000/tcp    eloquent_heisenberg
3dd7c49b5f08        training/webapp     "python app.py"     3 minutes ago       Up 3 minutes
      0.0.0.0:32768->5000/tcp   interesting_hawking
root@cg:~/course/docker/webapp/webapp#
```



查看端口

```
root@cg:~/course/docker/webapp/webapp# docker port 22cc30135359
5000/tcp -> 0.0.0.0:5000
root@cg:~/course/docker/webapp/webapp# docker port eloquent_heisenberg
5000/tcp -> 0.0.0.0:5000
root@cg:~/course/docker/webapp/webapp#
```

查看日志

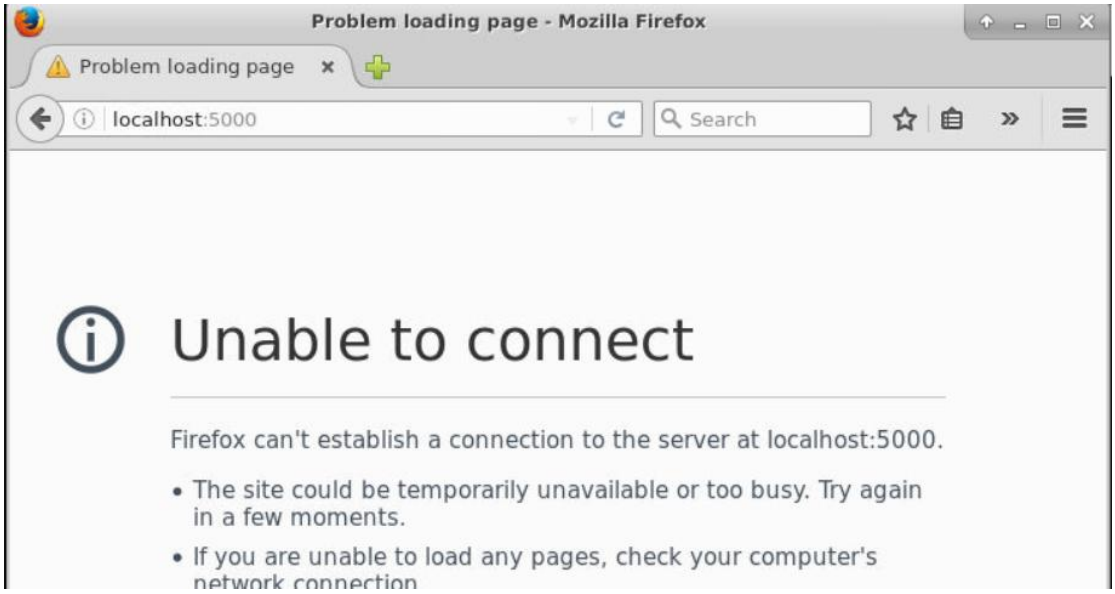
```
root@cg:~/course/docker/webapp/webapp# docker logs 22cc
 * Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
root@cg:~/course/docker/webapp/webapp# docker logs elo
Error: No such container: elo
root@cg:~/course/docker/webapp/webapp# docker logs eloquent_heisenberg
 * Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
root@cg:~/course/docker/webapp/webapp# docker logs -f eloquent_heisenberg
 * Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [09/Oct/2023 14:39:43] "GET /favicon.ico HTTP/1.1" 404 -
^C
```

查看 web 应用

```
root@cg:~/course/docker/webapp/webapp# docker top 22cc
UID                 PID                 PPID                C                   STIME
      TTY                 TIME                CMD
root                1883                1866                0                   14:39
      ?                   00:00:00            python app.py
root@cg:~/course/docker/webapp/webapp# docker inspect 22cc
[
    {
        "Id": "22cc30135359b35b818d5791d449f29118739b68ed2bd8332d1344b68b17dea3",
        "Created": "2023-10-09T14:39:00.424111897Z",
        "Path": "python",
        "Args": [
            "app.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 1883,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2023-10-09T14:39:01.245108354Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:6fae60ef344644649a39240b94d73b8ba9c67f898ede85cf8e947a887b3e6557",
        "ResolvConfPath": "/var/lib/docker/containers/22cc30135359b35b818d5791d449f29118739b6
```

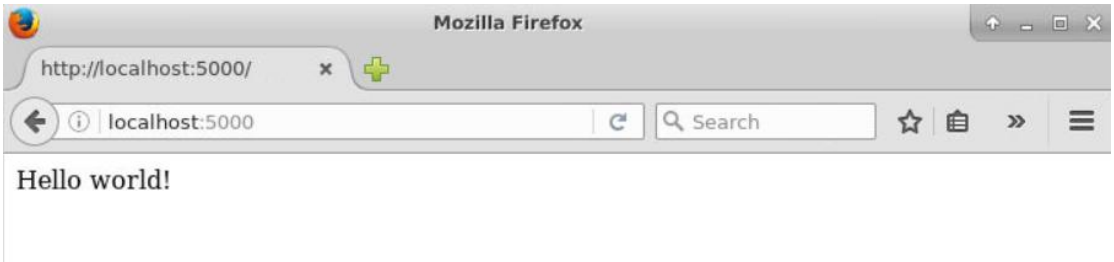
停止容器

```
root@cg:~/course/docker/webapp/webapp# docker stop 22cc
22cc
root@cg:~/course/docker/webapp/webapp# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
      PORTS                   NAMES
3dd7c49b5f08        training/webapp     "python app.py"     8 minutes ago       Up 8 minutes
      0.0.0.0:32768->5000/tcp   interesting_hawking
root@cg:~/course/docker/webapp/webapp# docker stop 3dd
3dd
root@cg:~/course/docker/webapp/webapp#
```



重启容器

```
root@cg:~/course/docker/webapp/webapp# docker start 22cc
22cc
root@cg:~/course/docker/webapp/webapp# docker [s
docker: '[s' is not a docker command.
See 'docker --help'
root@cg:~/course/docker/webapp/webapp# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
      PORTS                    NAMES
22cc30135359        training/webapp     "python app.py"     7 minutes ago       Up 9 seconds
      0.0.0.0:5000->5000/tcp   eloquent heisenberg
root@cg:~/course/docker/webapp/webapp#
```



移除容器

```
root@cg:~/course/docker/webapp/webapp# docker rm 22cc
Error response from daemon: You cannot remove a running container 22cc30135359b35b818d5791d449
f29118739b68ed2bd8332d1344b68b17dea3. Stop the container before attempting removal or force re
move
root@cg:~/course/docker/webapp/webapp# docker stop 22cc
22cc
root@cg:~/course/docker/webapp/webapp# docker rm 22cc
22cc
root@cg:~/course/docker/webapp/webapp#
```

实验 1.3

环境准备

```
root@cg:~# docker load < /headless/Desktop/images/ubuntu_15.10.tar.gz
f121afdbbd5d: Loading layer  142.9MB/142.9MB
4b955941a4d0: Loading layer  15.87kB/15.87kB
af288f00b8a7: Loading layer  11.78kB/11.78kB
98d59071f692: Loading layer  4.608kB/4.608kB
Loaded image: ubuntu:15.10
root@cg:~# docker load < /headless/Desktop/images/ubuntu_latest.tar.gz
ccdbb80308cc: Loading layer  75.07MB/75.07MB
63c99163f472: Loading layer  15.36kB/15.36kB
2f140462f3bc: Loading layer  3.584kB/3.584kB
Loaded image: ubuntu:latest
root@cg:~# docker load < /headless/Desktop/images/ubuntu_16.04.tar.gz
16012e0c1d35: Loading layer  138.2MB/138.2MB
31dda91e27a1: Loading layer  15.87kB/15.87kB
fbb91e31093e: Loading layer  11.78kB/11.78kB
1d5d9edd4bbc: Loading layer  3.072kB/3.072kB
Loaded image: ubuntu:16.04
root@cg:~# docker load < /headless/Desktop/images/hello-world_lastest.tar.gz
bash: /headless/Desktop/images/hello-world_lastest.tar.gz: No such file or direc
tory
root@cg:~# docker load < /headless/Desktop/images/hello-world_latest.tar.gz
f22b99068db9: Loading layer  15.36kB/15.36kB
Loaded image: hello-world:latest
root@cg:~# docker load < /headless/Desktop/images/training_webapp_latest.tar.gz
1154ba695078: Loading layer  197.2MB/197.2MB
528c8710fd95: Loading layer  208.9kB/208.9kB
37ee47034d9b: Loading layer  4.608kB/4.608kB
5f70bf18a086: Loading layer  1.024kB/1.024kB
b75c0703b86b: Loading layer  20.82MB/20.82MB
5c121779bb29: Loading layer  141.2MB/141.2MB
3ee91c5cb95b: Loading layer   2.56kB/2.56kB
6bbb1d0f8452: Loading layer  4.685MB/4.685MB
b44b0832d4c6: Loading layer  7.168kB/7.168kB
Loaded image: training/webapp:latest
root@cg:~#
```

查看镜像列表

```
root@cg:~# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
ubuntu              16.04               aefd7f02ae24        2 years ago         134MB
ubuntu              latest              7e0aa2d69a15        2 years ago         72.7MB
hello-world         latest              d1165f221234        2 years ago         13.3kB
ubuntu              15.10               9b9cb95443b5        7 years ago         137MB
training/webapp     latest              6fae60ef3446        8 years ago         349MB
```

获取新镜像

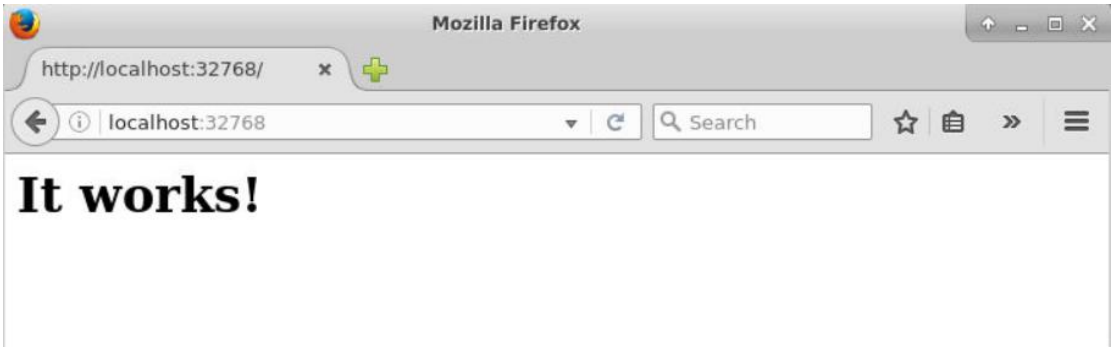
```
root@cg:~# docker load < /headless/Desktop/images/ubuntu_14.04.tar.gz
f2fa9f4cf8fd: Loading layer  206.1MB/206.1MB
30d3c4334a23: Loading layer  209.9kB/209.9kB
83109fa660b2: Loading layer  3.584kB/3.584kB
Loaded image: ubuntu:14.04
root@cg:~# docker run -i -t ubuntu:14.04 /bin/bash
root@686aae42662b:/# exit
exit
root@cg:~#
```

查找镜像

```
root@cg:~# docker search httpd
NAME                              DESCRIPTION
 STARS               OFFICIAL            AUTOMATED
httpd                             The Apache HTTP Server Project
 4559                [OK]
centos/httpd-24-centos7           Platform for running Apache httpd 2.4 or bui…
 46
centos/httpd
 36                                       [OK]
clearlinux/httpd                  httpd HyperText Transfer Protocol (HTTP) ser…
 5
centos/httpd-24-centos8
 3
solsson/httpd-openidc             mod_auth_openidc on official httpd image, ve…
 2                                        [OK]
hypoport/httpd-cgi                httpd-cgi
 2                                        [OK]
openquantumsafe/httpd             Demo of post-quantum cryptography in Apache …
 1
publici/httpd                     httpd:latest
 1                                        [OK]
dockerpinata/httpd
 1
dockette/apache                   Apache / HTTPD
 1                                        [OK]
manageiq/httpd                    Container with httpd, built on CentOS for Ma…
 1                                        [OK]
wodby/httpd
 0
jitesoft/httpd                    Apache httpd on Alpine linux.
 0
betterweb/httpd
 0
```

拖取镜像

```
root@cg:~# docker load < /headless/Desktop/images/httpd_latest.tar.gz
02c055ef67f5: Loading layer  72.53MB/72.53MB
15fd28211cd0: Loading layer  3.072kB/3.072kB
33c6c92714e0: Loading layer  7.483MB/7.483MB
a5c7a08a53e0: Loading layer  61.53MB/61.53MB
7e462892f8e9: Loading layer  3.584kB/3.584kB
Loaded image: httpd:latest
root@cg:~# docker run -itd -P --name httpd_test httpd
138d4af0987c2e02c31fd4a1bd529ded6072fbcf3aaf56ce4ad6ba7746d33d1b
root@cg:~# docker ps -l
CONTAINER ID        IMAGE               COMMAND             CREATED
 STATUS              PORTS                    NAMES
138d4af0987c        httpd               "httpd-foreground"   6 seconds ago
 Up 5 seconds        0.0.0.0:32768->80/tcp   httpd_test
root@cg:~#
```



删除镜像

```
root@cg:~# docker rmi hello-world
Untagged: hello-world:latest
Deleted: sha256:d1165f2212346b2bab48cb01c1e39ee8ad1be46b87873d9ca7a4e434980a7726
Deleted: sha256:f22b99068db93900abe17f7f5e09ec775c2826ecfe9db961fea68293744144bd
root@cg:~# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED
SIZE
httpd               latest              f3cffeea581b        2 years ago
138MB
ubuntu              16.04               aefd7f02ae24        2 years ago
134MB
ubuntu              latest              7e0aa2d69a15        2 years ago
72.7MB
ubuntu              14.04               13b66b487594        2 years ago
197MB
ubuntu              15.10               9b9cb95443b5        7 years ago
137MB
training/webapp     latest              6fae60ef3446        8 years ago
349MB
root@cg:~#
```

更新 apt 工具

```
root@b4a828c8b5f4:/# apt-get update
Get:1 http://security.ubuntu.com/ubuntu xenial-security InRelease [99.8 kB]
Get:2 http://archive.ubuntu.com/ubuntu xenial InRelease [247 kB]
Get:3 http://archive.ubuntu.com/ubuntu xenial-updates InRelease [99.8 kB]
Get:4 http://archive.ubuntu.com/ubuntu xenial-backports InRelease [97.4 kB]
Get:5 http://archive.ubuntu.com/ubuntu xenial/main amd64 Packages [1558 kB]
Get:6 http://archive.ubuntu.com/ubuntu xenial/restricted amd64 Packages [14.1 kB]
Get:7 http://archive.ubuntu.com/ubuntu xenial/universe amd64 Packages [9827 kB]
Get:8 http://archive.ubuntu.com/ubuntu xenial/multiverse amd64 Packages [176 kB]
Get:9 http://archive.ubuntu.com/ubuntu xenial-updates/main amd64 Packages [2560 kB]
Get:10 http://archive.ubuntu.com/ubuntu xenial-updates/restricted amd64 Packages [16.4 kB
Get:11 http://archive.ubuntu.com/ubuntu xenial-updates/universe amd64 Packages [1545 kB]
Get:12 http://archive.ubuntu.com/ubuntu xenial-updates/multiverse amd64 Packages [25.0 kB
Get:13 http://archive.ubuntu.com/ubuntu xenial-backports/main amd64 Packages [10.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu xenial-backports/universe amd64 Packages [12.7 kB
Get:1 http://security.ubuntu.com/ubuntu xenial-security InRelease [99.8 kB]
Get:15 http://security.ubuntu.com/ubuntu xenial-security/main amd64 Packages [2052 kB]
```

安装 gcc

```
root@b4a828c8b5f4:/# apt-get install gcc
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  binutils cpp cpp-5 gcc-5 libasan2 libatomic1 libc-dev-bin libc6 libc6-dev libcc1-0
  libcilkrts5 libgcc-5-dev libgmp10 libgomp1 libisl15 libitm1 liblsan0 libmpc3 libmpfr4
  libmpx0 libquadmath0 libtsan0 libubsan0 linux-libc-dev manpages manpages-dev
Suggested packages:
  binutils-doc cpp-doc gcc-5-locales gcc-multilib make autoconf automake libtool flex bison
  gdb gcc-doc gcc-5-multilib gcc-5-doc libgcc1-dbg libgomp1-dbg libitm1-dbg libatomic1-dbg
  libasan2-dbg liblsan0-dbg libtsan0-dbg libubsan0-dbg libcilkrts5-dbg libmpx0-dbg
  libquadmath0-dbg glibc-doc locales man-browser
The following NEW packages will be installed:
  binutils cpp cpp-5 gcc gcc-5 libasan2 libatomic1 libc-dev-bin libc6-dev libcc1-0
  libcilkrts5 libgcc-5-dev libgmp10 libgomp1 libisl15 libitm1 liblsan0 libmpc3 libmpfr4
  libmpx0 libquadmath0 libtsan0 libubsan0 linux-libc-dev manpages manpages-dev
The following packages will be upgraded:
  libc6
1 upgraded, 26 newly installed, 0 to remove and 6 not upgraded.
Need to get 31.7 MB of archives.
After this operation, 103 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu xenial-updates/main amd64 libc6 amd64 2.23-0ubuntu11.3
[2590 kB]
Get:2 http://archive.ubuntu.com/ubuntu xenial/main amd64 libgmp10 amd64 2:6.1.0+dfsg-2 [240 kB
]
Get:3 http://archive.ubuntu.com/ubuntu xenial/main amd64 libmpfr4 amd64 3.1.4-1 [191 kB]
Get:4 http://archive.ubuntu.com/ubuntu xenial/main amd64 libmpc3 amd64 1.0.3-1 [39.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu xenial/main amd64 manpages all 4.04-2 [1087 kB]
Get:6 http://archive.ubuntu.com/ubuntu xenial-updates/main amd64 binutils amd64 2.26.1-1ubuntu
```

```
root@b4a828c8b5f4:/# gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/lib/gcc/x86_64-linux-gnu/5/lto-wrapper
Target: x86_64-linux-gnu
Configured with: ../src/configure -v --with-pkgversion='Ubuntu 5.4.0-6ubuntu1~16.04.12' --with
-bugurl=file:///usr/share/doc/gcc-5/README.Bugs --enable-languages=c,ada,c++,java,go,d,fortran
,objc,obj-c++ --prefix=/usr --program-suffix=-5 --enable-shared --enable-linker-build-id --lib
execdir=/usr/lib --without-included-gettext --enable-threads=posix --libdir=/usr/lib --enable-
nls --with-sysroot=/ --enable-clocale=gnu --enable-libstdcxx-debug --enable-libstdcxx-time=yes
 --with-default-libstdcxx-abi=new --enable-gnu-unique-object --disable-vtable-verify --enable-
libmpx --enable-plugin --with-system-zlib --disable-browser-plugin --enable-java-awt=gtk --ena
ble-gtk-cairo --with-java-home=/usr/lib/jvm/java-1.5.0-gcj-5-amd64/jre --enable-java-home --wi
th-jvm-root-dir=/usr/lib/jvm/java-1.5.0-gcj-5-amd64 --with-jvm-jar-dir=/usr/lib/jvm-exports/ja
va-1.5.0-gcj-5-amd64 --with-arch-directory=amd64 --with-ecj-jar=/usr/share/java/eclipse-ecj.ja
r --enable-objc-gc --enable-multiarch --disable-werror --with-arch-32=i686 --with-abi=m64 --wi
th-multilib-list=m32,m64,mx32 --enable-multilib --with-tune=generic --enable-checking=release
--build=x86_64-linux-gnu --host=x86_64-linux-gnu --target=x86_64-linux-gnu
Thread model: posix
gcc version 5.4.0 20160609 (Ubuntu 5.4.0-6ubuntu1~16.04.12)
root@b4a828c8b5f4:/#
```

生成新镜像

```
root@cg:~# docker commit -m "gcc installed" -a "cg" b4a828c8b5f4 cg/my_ubuntu:v2.0
sha256:4243dfb9949432e095a5f0135c2027f7cc8b73eaa9f6f7279d7d53d1d89f6b85
root@cg:~# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
cg/my_ubuntu        v2.0                4243dfb99494        6 seconds ago       275MB
httpd               latest              f3cffeea581b        2 years ago         138MB
ubuntu              16.04               aefd7f02ae24        2 years ago         134MB
ubuntu              latest              7e0aa2d69a15        2 years ago         72.7MB
ubuntu              14.04               13b66b487594        2 years ago         197MB
ubuntu              15.10               9b9cb95443b5        7 years ago         137MB
training/webapp     latest              6fae60ef3446        8 years ago         349MB
root@cg:~#
```

创建镜像，编译并运行 c 语言程序

```
root@cg:~/course/docker/c_hello_world# docker run -itd --name ubuntu_gcc cg/my_ubuntu:v2.0
9ef9210cf7c7217e9be696d962ba7632f7ccf8e23622d4213e03c31a47790ba8
root@cg:~/course/docker/c_hello_world# docker ps -l
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
      PORTS               NAMES
9ef9210cf7c7        cg/my_ubuntu:v2.0   "/bin/bash"         7 seconds ago       Up 5 seconds
                          ubuntu_gcc
```

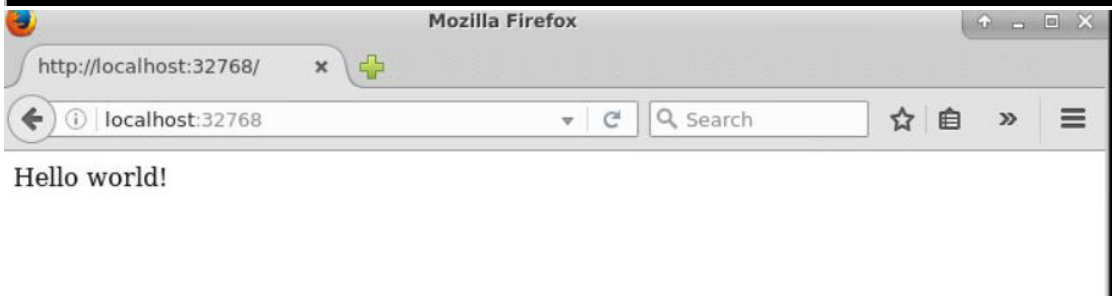
```
root@cg:~/course/docker/c_hello_world# docker cp hello_world.c 9ef9210cf7c7:/home
root@cg:~/course/docker/c_hello_world# docker exec -it 9ef9 /bin/bash
root@9ef9210cf7c7:/# cd home
root@9ef9210cf7c7:/home# ls -l
total 4
-rw-r--r-- 1 root root 72 Oct 10 08:29 hello_world.c
root@9ef9210cf7c7:/home# gcc hello_world.c -o hello_world
root@9ef9210cf7c7:/home# ./hello_world
Hello World!
root@9ef9210cf7c7:/home# exit
exit
```

设置镜像标签

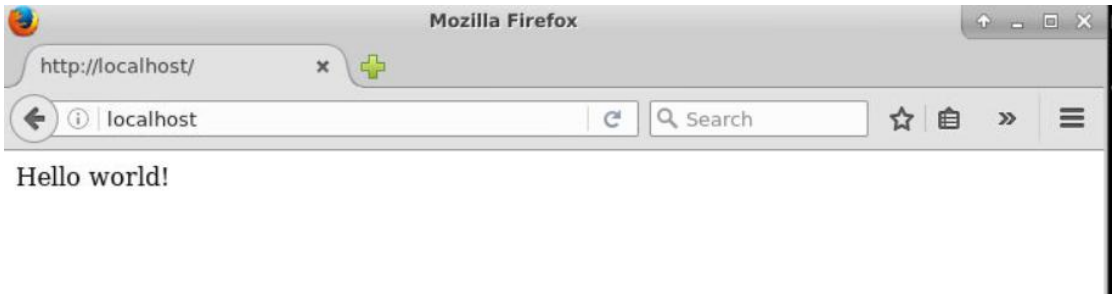
```
root@cg:~/course/docker/c_hello_world# docker tag 4243 cg/ubuntu_gcc:dec
root@cg:~/course/docker/c_hello_world# docker tag 4243 cg/ubuntu_gcc:dev
root@cg:~/course/docker/c_hello_world# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
cg/my_ubuntu        v2.0                4243dfb99494        17 minutes ago      275MB
cg/ubuntu_gcc       dec                 4243dfb99494        17 minutes ago      275MB
cg/ubuntu_gcc       dev                 4243dfb99494        17 minutes ago      275MB
httpd               latest              f3cffeea581b        2 years ago         138MB
ubuntu              16.04               aefd7f02ae24        2 years ago         134MB
ubuntu              latest              7e0aa2d69a15        2 years ago         72.7MB
ubuntu              14.04               13b66b487594        2 years ago         197MB
ubuntu              15.10               9b9cb95443b5        7 years ago         137MB
training/webapp     latest              6fae60ef3446        8 years ago         349MB
root@cg:~/course/docker/c_hello_world#
```

实验 1.4

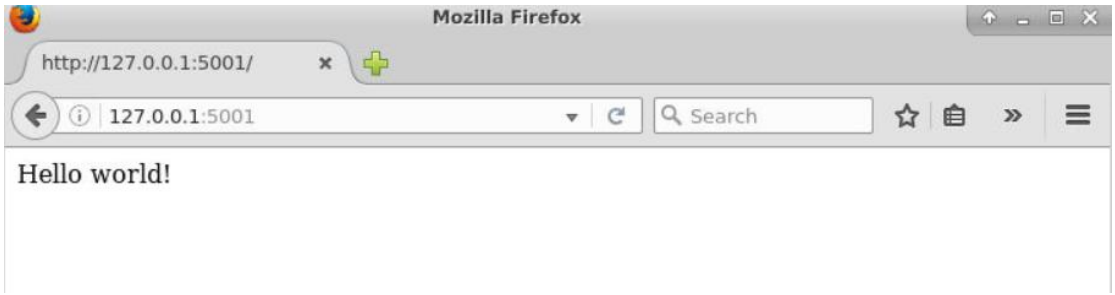
```
root@cg:~# cd course/docker/webapp/webapp
root@cg:~/course/docker/webapp/webapp# docker run -d -P training/webapp python a
pp.py
Unable to find image 'training/webapp:latest' locally
latest: Pulling from training/webapp
Image docker.io/training/webapp:latest uses outdated schema1 manifest format. Pl
ease upgrade to a schema2 image for better future compatibility. More informatio
n at https://docs.docker.com/registry/spec/deprecated-schema-v1/
e190868d63f8: Pull complete
909cd34c6fd7: Pull complete
0b9bfabab7c1: Pull complete
a3ed95caeb02: Pull complete
10bbbc0fc0ff: Pull complete
fca59b508e9f: Pull complete
e7ae2541b15b: Pull complete
9dd97ef58ce9: Pull complete
a4c1b0cb7af7: Pull complete
Digest: sha256:06e9c1983bd6d5db5fba376ccd63bfa529e8d02f23d5079b8f74a616308fb11d
Status: Downloaded newer image for training/webapp:latest
3fcadc5589dbf6948a5515601f472396df4183880b487b7ed03c8c3623ae0c49
root@cg:~/course/docker/webapp/webapp# docker ps -l
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS              PORTS                   NAMES
3fcadc5589db        training/webapp     "python app.py"     20 seconds ago
Up 17 seconds       0.0.0.0:32768->5000/tcp   elegant_chebyshev
root@cg:~/course/docker/webapp/webapp#
```




```
root@cg:~/course/docker/webapp/webapp# docker run -d -p 80:5000 training/webapp python
1c4d1d0163103743a08bef9fe6d0ef12ec98af20957b864b1b0b1a5235b4fd55
root@cg:~/course/docker/webapp/webapp#
```




```
root@cg:~/course/docker/webapp/webapp# docker run -d -p 127.0.0.1:5001:5000 training/web
thon app.py
00e3084d6248ba599b328351614f1a479be5a15872ea2f2892728833c1df5a33
```




```
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the
dule) (@INC contains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.30.0 /
erl/5.30.0 /usr/lib/x86_64-linux-gnu/perl5/5.30 /usr/share/perl5 /usr/lib/x86
l/5.30 /usr/share/perl/5.30 /usr/local/lib/site_perl /usr/lib/x86_64-linux-gn
usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Updating certificates in /etc/ssl/certs...
137 added, 0 removed; done.
Processing triggers for libc-bin (2.31-0ubuntu9.2) ...
Processing triggers for ca-certificates (20230311ubuntu0.20.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@ed8ffacaaf3f:/# ping 172.17.0.7
bash: ping: command not found
root@ed8ffacaaf3f:/# wget 172.17.0.7
--2023-10-10 09:04:05--  http://172.17.0.7/
Connecting to 172.17.0.7:80... failed: No route to host.
root@ed8ffacaaf3f:/# wget 127.0.0.1
--2023-10-10 09:07:36--  http://127.0.0.1/
Connecting to 127.0.0.1:80... failed: Connection refused.
root@ed8ffacaaf3f:/# wget 127.0.0.1:5001
--2023-10-10 09:07:45--  http://127.0.0.1:5001/
Connecting to 127.0.0.1:5001... failed: Connection refused.
root@ed8ffacaaf3f:/# wget 127.0.0.1:32768
--2023-10-10 09:07:59--  http://127.0.0.1:32768/
Connecting to 127.0.0.1:32768... failed: Connection refused.
root@ed8ffacaaf3f:/#
```

使用 link 进行容器互连

```
root@cg:~# docker run -d --name web training/webapp python app.py
d4c0795d434434f97c1725745c607f3ed6d06caa4db548ebafefbd76f44a8ed5
root@cg:~# docker run -itd --link web:pyweb ubuntu
ddbbfd675f8eff80121f3753e800b511065f9006d2ba143b677b9b968e2c0419
root@cg:~# docker ps -l
CONTAINER ID        IMAGE               COMMAND             CREATED
STATUS              PORTS               NAMES
ddbbfd675f8e        ubuntu              "/bin/bash"         6 seconds ago
Up 4 seconds                            stoic_bell
root@cg:~# docker exec -it ddbb /bin/bash
root@ddbbfd675f8e:/# apt-get update
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages
 [29.3 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3109
```

```
root@ddbbfd675f8e:/# wget pyweb:5000
--2023-10-10 09:27:39--  http://pyweb:5000/
Resolving pyweb (pyweb)... 172.17.0.2
Connecting to pyweb (pyweb)|172.17.0.2|:5000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12 [text/html]
Saving to: 'index.html'

index.html          100%[===================>]      12  --.-KB/s    in 0s

2023-10-10 09:27:39 (1.33 MB/s) - 'index.html' saved [12/12]

root@ddbbfd675f8e:/# cat index.html
Hello world!root@ddbbfd675f8e:/#
```

查看环境变量

```
root@cg:~# docker run --rm --name test2 --link web:pyweb ubuntu env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=8f755f561ea6
PYWEB_PORT=tcp://172.17.0.2:5000
PYWEB_PORT_5000_TCP=tcp://172.17.0.2:5000
PYWEB_PORT_5000_TCP_ADDR=172.17.0.2
PYWEB_PORT_5000_TCP_PORT=5000
PYWEB_PORT_5000_TCP_PROTO=tcp
PYWEB_NAME=/test2/pyweb
HOME=/root
root@cg:~# docker run --rm --name test2 --link web:pyweb ubuntu cat /etc/hosts
127.0.0.1       localhost
::1     localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.17.0.2      pyweb d4c0795d4344 web
172.17.0.4      d3e55f349961
root@cg:~#
```

新建网络

```
root@cg:~# docker network create -d bridge mynet
76e0bf4446b735a68b975be276f6cd0387e81c2ee1d34e4555cbeb38e7cd7c7a
root@cg:~# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
7ed5fa26a77a        bridge              bridge              local
eabe6756e128        host                host                local
76e0bf4446b7        mynet               bridge              local
aab0582e3a6b        none                null                local
root@cg:~#
```

```
root@cg:~# docker run -itd --name test1 --network mynet ubuntu
b57889ec2901246a89056fe8b6fc257914004d25f1165257a22c30bbc66f66d3
root@cg:~# docker run -itd --name test2 --network mynet ubuntu
7b4397e6da1583efbfcfba33a699754b4e1edc0a24b23f9e475cce16690a816a
root@cg:~# docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
      PORTS               NAMES
7b4397e6da15        ubuntu              "/bin/bash"         3 seconds ago       Up 3 seconds
                          test2
b57889ec2901        ubuntu              "/bin/bash"         16 seconds ago      Up 15 seconds
                          test1
ddbbfd675f8e        ubuntu              "/bin/bash"         10 minutes ago      Up 10 minutes
                          stoic_bell
d4c0795d4344        training/webapp     "python app.py"     10 minutes ago      Up 10 minutes
      5000/tcp            web
root@cg:~#
```

Ping 命令检验连接

```
root@cg:~# docker run -itd --name test1 --network mynet cg/ubuntu_net:v1.0
9a7f8ba4e4c17ccabe67206d83c2ebbd4cc95829004dce85ebe44adae62b3a2e
root@cg:~# docker run -itd --name test2 --network mynet cg/ubuntu_net:v1.0
c045ca097ff82964c07215774c7bac0a68bd7bfd8fec039d38f4a001995d71d4
root@cg:~# docker exec -it test1 /bin/bash
root@9a7f8ba4e4c1:/# pint test2
bash: pint: command not found
root@9a7f8ba4e4c1:/# ping test2
PING test2 (172.18.0.4) 56(84) bytes of data.
64 bytes from test2.mynet (172.18.0.4): icmp_seq=1 ttl=64 time=0.141 ms
64 bytes from test2.mynet (172.18.0.4): icmp_seq=2 ttl=64 time=0.038 ms
64 bytes from test2.mynet (172.18.0.4): icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from test2.mynet (172.18.0.4): icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from test2.mynet (172.18.0.4): icmp_seq=5 ttl=64 time=0.032 ms
^C
--- test2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4081ms
rtt min/avg/max/mdev = 0.030/0.055/0.141/0.042 ms
root@9a7f8ba4e4c1:/#
root@9a7f8ba4e4c1:/# exit
exit
root@cg:~# docker exec -it test2 /bin/bash
root@c045ca097ff8:/# ping test1
PING test1 (172.18.0.3) 56(84) bytes of data.
64 bytes from test1.mynet (172.18.0.3): icmp_seq=1 ttl=64 time=0.101 ms
64 bytes from test1.mynet (172.18.0.3): icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from test1.mynet (172.18.0.3): icmp_seq=3 ttl=64 time=0.042 ms
64 bytes from test1.mynet (172.18.0.3): icmp_seq=4 ttl=64 time=0.035 ms
64 bytes from test1.mynet (172.18.0.3): icmp_seq=5 ttl=64 time=0.038 ms
^C
--- test1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4084ms
rtt min/avg/max/mdev = 0.035/0.054/0.101/0.024 ms
root@c045ca097ff8:/#
```

指定 ip

```
root@cg:~# docker network create --subnet=172.19.0.0/16 mynet
978d80a4c335298cfcf5642224a2a2d1068d42ae4507de89cfc37c7035760bae
root@cg:~# docker run -itd --name test1 --network mynet --ip 172.19.1.1 cg/ubunt
u_net:v1.0
49d8f248d03cb22c74be9fbf3ed50ec21ac1ee2b2df06dcec3110b3070f6747f
root@cg:~# docker run -itd --name test2 --network mynet --ip 172.19.1.1 cg/ubunt
u_net:v1.0
306880b3952a40cbae0fb16d8b296ad9ed4ab2fbf97cd5570090ad15f841c017
docker: Error response from daemon: Address already in use.
root@cg:~# docker run -itd --name test2 --network mynet --ip 172.19.1.2 cg/ubunt
u_net:v1.0
docker: Error response from daemon: Conflict. The container name "/test2" is alr
eady in use by container "306880b3952a40cbae0fb16d8b296ad9ed4ab2fbf97cd5570090ad
15f841c017". You have to remove (or rename) that container to be able to reuse t
hat name.
See 'docker run --help'.
root@cg:~# docker stop test2
test2
root@cg:~# docker rm test2
test2
root@cg:~# docker run -itd --name test2 --network mynet --ip 172.19.1.2 cg/ubunt
u_net:v1.0
c03aa248e032dd52076aa1bab74f9bfb935a7e159300dbe6d996466d0279a085
root@cg:~# docker exec -it test1 /bin/bash
root@49d8f248d03c:/# hostname -I
172.19.1.1
root@49d8f248d03c:/# ping -c 3 test2
PING test2 (172.19.1.2) 56(84) bytes of data.
64 bytes from test2.mynet (172.19.1.2): icmp_seq=1 ttl=64 time=0.159 ms
64 bytes from test2.mynet (172.19.1.2): icmp_seq=2 ttl=64 time=0.045 ms
64 bytes from test2.mynet (172.19.1.2): icmp_seq=3 ttl=64 time=0.027 ms

--- test2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2041ms
```

```
root@cg:~# docker exec -it test2 /bin/bash
root@c03aa248e032:/# hostname -I
172.19.1.2
root@c03aa248e032:/# ping -c 3 tets1
ping: tets1: Name or service not known
root@c03aa248e032:/# ping -c 3 test1
PING test1 (172.19.1.1) 56(84) bytes of data.
64 bytes from test1.mynet (172.19.1.1): icmp_seq=1 ttl=64 time=0.090 ms
64 bytes from test1.mynet (172.19.1.1): icmp_seq=2 ttl=64 time=0.033 ms
64 bytes from test1.mynet (172.19.1.1): icmp_seq=3 ttl=64 time=0.024 ms

--- test1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2054ms
rtt min/avg/max/mdev = 0.024/0.049/0.090/0.029 ms
root@c03aa248e032:/#
```

```
--- test1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2054ms
rtt min/avg/max/mdev = 0.024/0.049/0.090/0.029 ms
root@c03aa248e032:/# exit
exit
root@cg:~# docker run -it --rm --hostname=ubuntu_test cg/ubuntu_net:v1.0 cat /etc/hosts
127.0.0.1       localhost
::1     localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.17.0.2      ubuntu_test
root@cg:~# docker run -it --rm --dns=114.114.114.114 cg/ubuntu_net:v1.0 cat /etc/resolv.confse
arch int.educg.net
cat: /etc/resolv.confsearch: No such file or directory
cat: int.educg.net: No such file or directory
root@cg:~# docker run -it --rm --dns=114.114.114.114 cg/ubuntu_net:v1.0 cat /etc/resolv.confse
arch int.educg.net
cat: /etc/resolv.confsearch: No such file or directory
cat: int.educg.net: No such file or directory
root@cg:~#
```